

E-Commerce Hybrid Recommendation System

Olist Brazilian E-Commerce Dataset | Veri Bilimi Final Project | Ahmet Bilal Ozgun

1. Introduction and Motivation

This project builds and evaluates a hybrid recommendation system for the Olist Brazilian E-Commerce dataset. The practical goal is to combine observed customer behavior with product metadata so that recommendations can reflect both preference history and semantic product similarity. The project studies three questions: whether multi-category buyers rate products differently by region, whether category-level SVD improves over a global-mean baseline, and whether semantic product similarity aligns with real co-purchase behavior.

2. Dataset and Cleaning Decisions

The analysis uses the public Olist marketplace data from Kaggle under the CC BY-NC-SA 4.0 license. The raw data contains orders, order items, products, customers, reviews, payments, sellers, geolocation records, and category translations. The modeling dataset keeps only delivered orders because cancelled, unavailable, or in-transit orders do not represent complete purchase experiences with reliable review signals.

Step	Result
Raw orders	99,441
Delivered orders	96,478
Products after null-category drop	32,341
Final master table	108,660 rows x 29 columns
Mean review score	4.157

Rows with missing product categories were dropped because category is required by both the collaborative-filtering matrix and content metadata strings. Payment rows were aggregated to one row per order to avoid join fan-out, while duplicate review rows were resolved by keeping the latest answer timestamp. The resulting master table is item-level, so order-level summaries deduplicate by order_id.

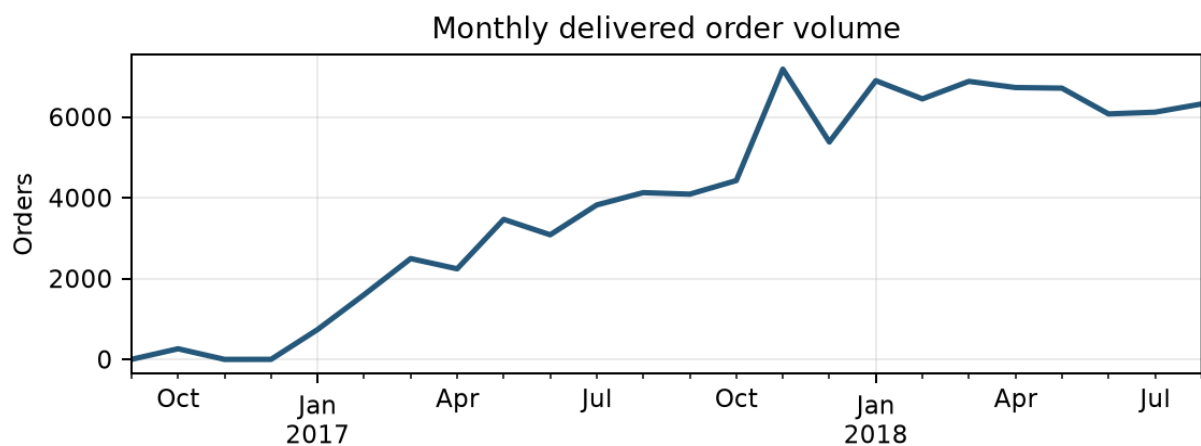


Figure 1. Delivered orders peak in 2017-11 with 7,187 orders.

3. Methods

Collaborative filtering uses scikit-surprise SVD on a customer_id by product_category_name matrix, where each value is the mean review score for that customer-category pair. Category granularity is intentional: product_id-level modeling is too sparse for this dataset. Evaluation uses temporal splits rather than random splits, preventing future reviews from leaking into training.

Content-based filtering represents each product by a short metadata string containing category, price bucket, and weight bucket. Sentence-transformer embeddings are L2-normalized and indexed with FAISS IndexFlatIP, so inner product equals cosine similarity. The hybrid demo combines normalized SVD predictions and content similarity as $\text{hybrid_score} = 0.6 * \text{CF} + 0.4 * \text{CB}$, giving slightly more weight to observed behavior while retaining semantic fallback value.

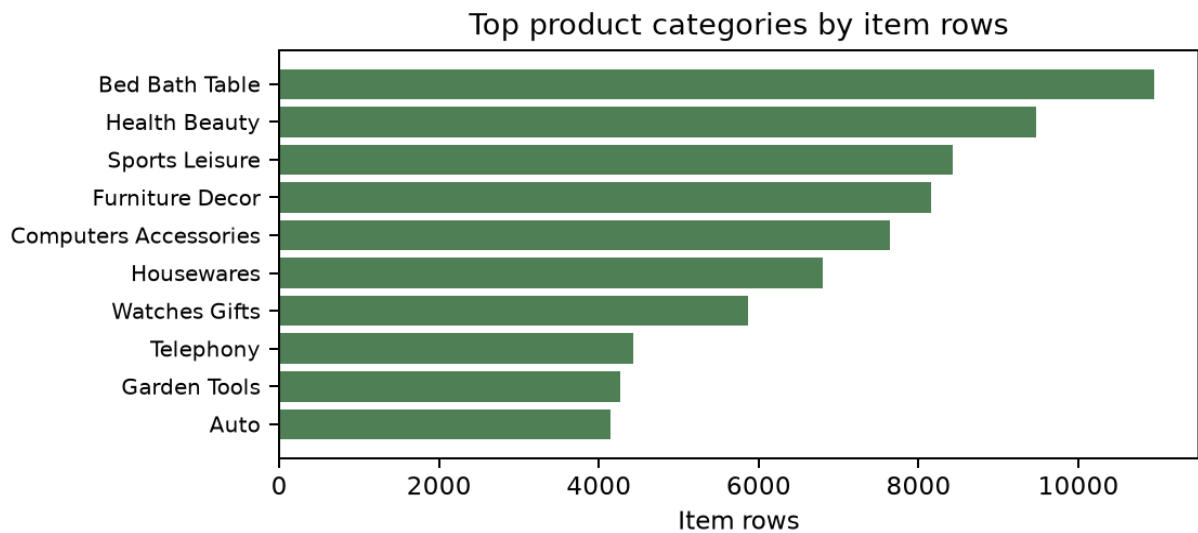


Figure 2. Category distribution is highly concentrated in a small number of categories.

4. Results

RQ1 asks whether multi-category buyers rate products differently by Brazilian region. Across the top five states, multi-category buyers have lower mean review scores than single-category buyers. The Mann-Whitney U test is statistically significant, but the effect size is negligible, so the practical difference is small despite the large sample.

Metric	Value
U statistic	83,140,360
p-value	4.567e-34
n multi-category	2,154
n single-category	89,347
Effect size r	0.0357
Mean score, multi vs single	3.886 vs 4.161



Figure 3. Multi-category customers rate slightly lower in each of the five largest states.

4. Results, Continued

RQ2 evaluates whether SVD improves over a global-mean baseline. The matrix contains 95,242 customer-category pairs for 94,508 users and 73 categories, with sparsity 0.9862. Because most Olist customers buy once, the post-2018-01 temporal test split is entirely cold-start at the customer level. SVD therefore behaves close to the global mean, producing RMSE 1.3113 versus 1.3137, an improvement of 0.18%.

Split / Metric	SVD RMSE	Global RMSE	Improvement
Primary: train pre-2018-01, test 2018-01+	1.3113	1.3137	0.18%
2017-07 to 2017-10	1.1741	1.1745	0.04%
2017-10 to 2018-01	1.3441	1.3492	0.38%
2018-01 to 2018-04	1.4579	1.4614	0.24%

Across the three temporal validation windows, mean SVD RMSE is 1.3254 with standard deviation 0.1429. The result is academically useful because it demonstrates the main limitation of this marketplace setting: sparse repeat-user history limits personalization.

RQ3 compares FAISS semantic neighbors to products that appear in the same order. The result is mean overlap@5 = 0.000 / 5 over 49 high-volume products. This near-zero overlap is expected because 90.1% of orders contain one item and 96.7% contain one unique product after collapsing repeated quantities. In this dataset, co-purchase overlap is too sparse to be a strong content-based evaluation signal.

RQ3 Metric	Value
Products evaluated	49
Mean overlap@5	0.000 / 5
Products with non-empty co-purchase set	45 / 49
Single-item orders	90.1%
Single-unique-product orders	96.7%

5. Conclusions and Limitations

The hybrid approach is appropriate for the project because it combines behavioral preference learning with semantic similarity, but the data strongly constrains the measurable gains. The regional analysis finds a statistically significant but practically tiny rating difference for multi-category buyers. The SVD model barely improves over the global mean because the temporal test set is dominated by cold-start customers. The content-based system produces coherent metadata neighbors, but co-purchase validation is weak because most orders contain only one product.

Main limitations are cold-start customers, high matrix sparsity, category-level rather than product-level collaborative filtering, short product metadata strings, and no recency weighting. Future work should add richer product text, repeat-buyer segmentation, and a ranking metric based on held-out future purchases when enough user history is available.